



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

DISCRETE-TIME SIGNALS AND SYSTEMS



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OBJECTIVES

- Understand the discrete-time signals and systems.
- Know the way to show the discrete-time signals and systems.
- Understand the Fourier transform.

1. Discrete-time signals
2. Discrete-time systems
3. The Linear Time Invariant (LTI) systems
4. Represent the discrete-time signals & systems in the frequency domain
5. Represent the sequences by Fourier transform
6. Symmetry properties of the Fourier transform
7. Fourier transform theorems
8. Discrete-time random signals

- Discrete-time signals are represented as

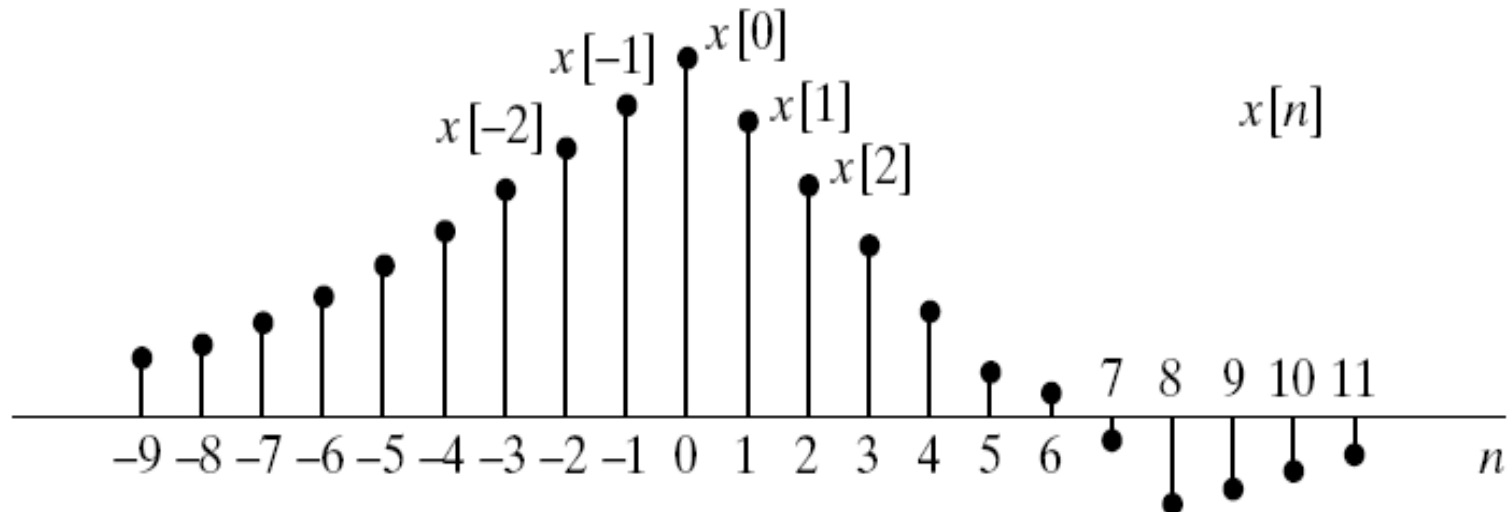
$$x = \{x[n]\}, \quad -\infty < n < \infty, \quad n : \text{integer}$$

Cumbersome, so just use $x[n]$

- In sampling,

$$x[n] = x_a(nT), \quad T : \text{sampling period}$$

- $1/T$ (reciprocal of T): sampling frequency



Graphical representation of a discrete-time signals

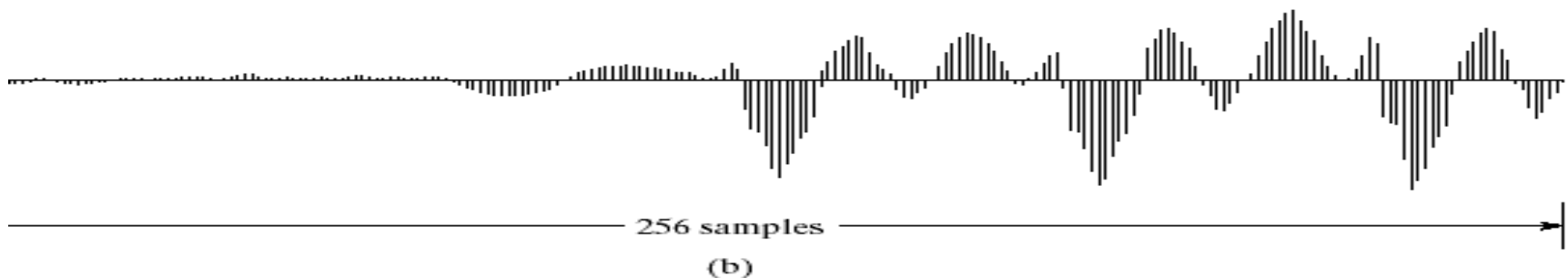
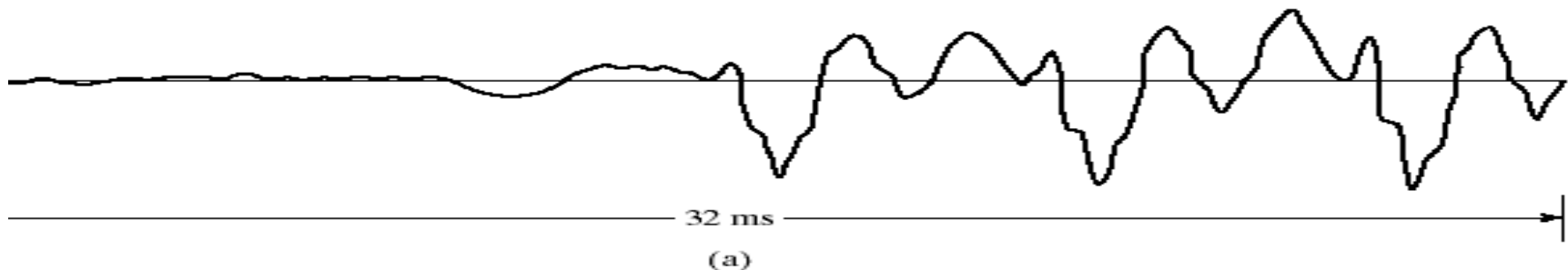
Abscissa: continuous line

$x[n]$: is defined only at discrete instants

$$x[n] = x_a(t) \big|_{t=nT} = x_a(nT)$$

EXAMPLE

Sampling the analog waveform



- Basic Sequence Operations

- Sum of two sequences: $x[n] + y[n]$

- Product of two sequences: $x[n] \cdot y[n]$

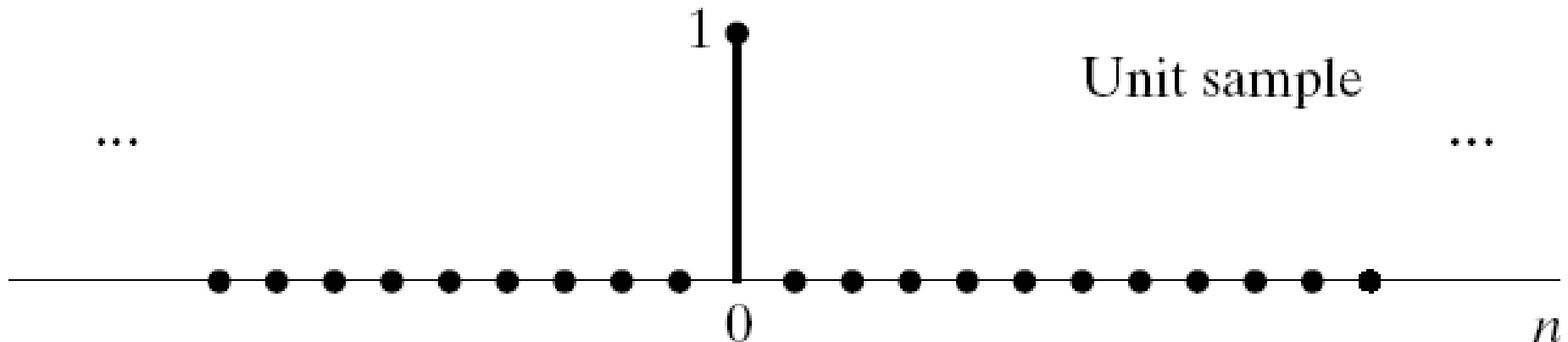
- Multiplication of a sequence by a number $\alpha \cdot x[n]$

- Delay (shift) of a sequence

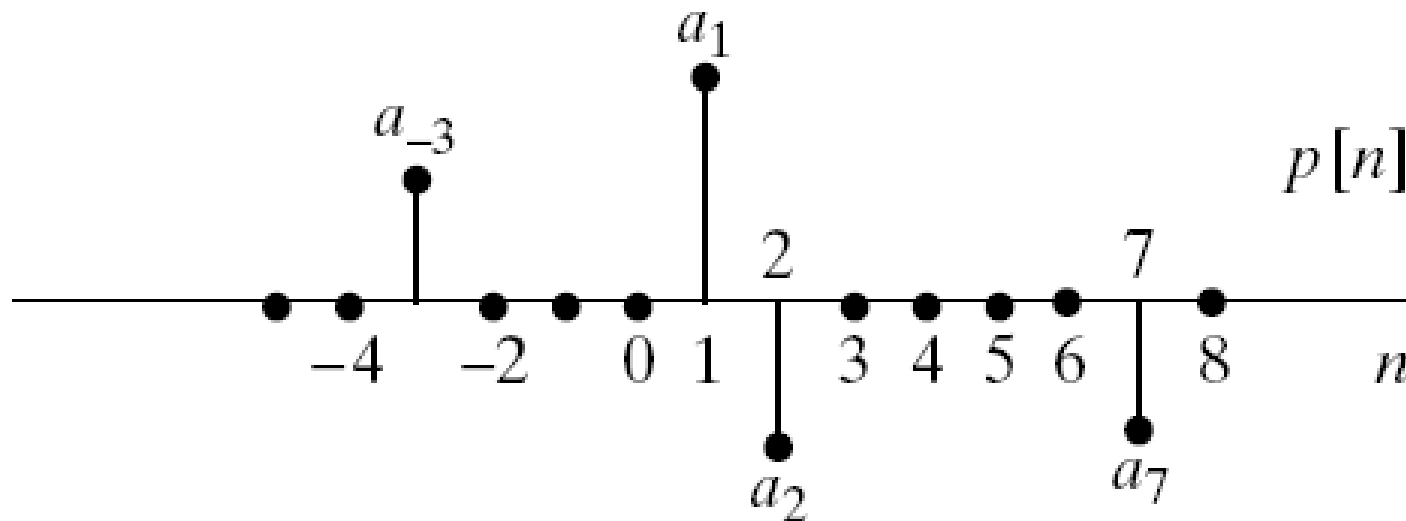
$$y[n] = x[n - n_0] \quad n_0 : \text{integer}$$

- Basic sequences
 - Unit sample sequence (discrete-time impulse, impulse)

$$\delta[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n = 0 \end{cases}$$



EXAMPLE : A sum of scaled, delayed impulses



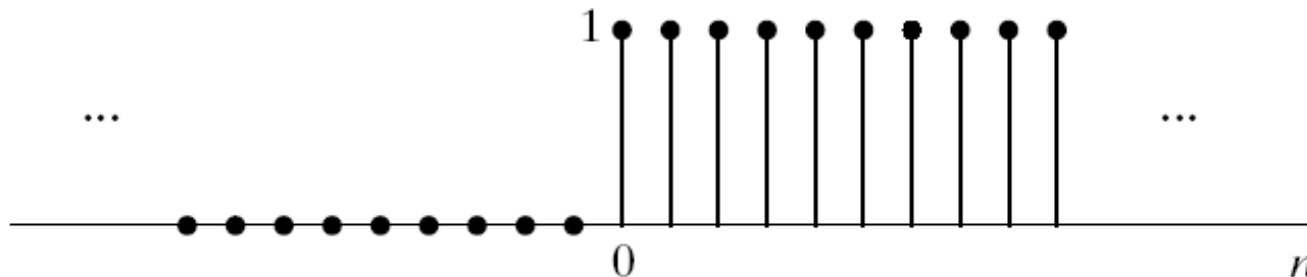
$$p[n] = a_{-3}\delta[n+3] + a_1\delta[n-1] + a_2\delta[n-2] + a_7\delta[n-7]$$

◆ **Arbitrary sequence** $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$

• Unit step sequence

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Unit step



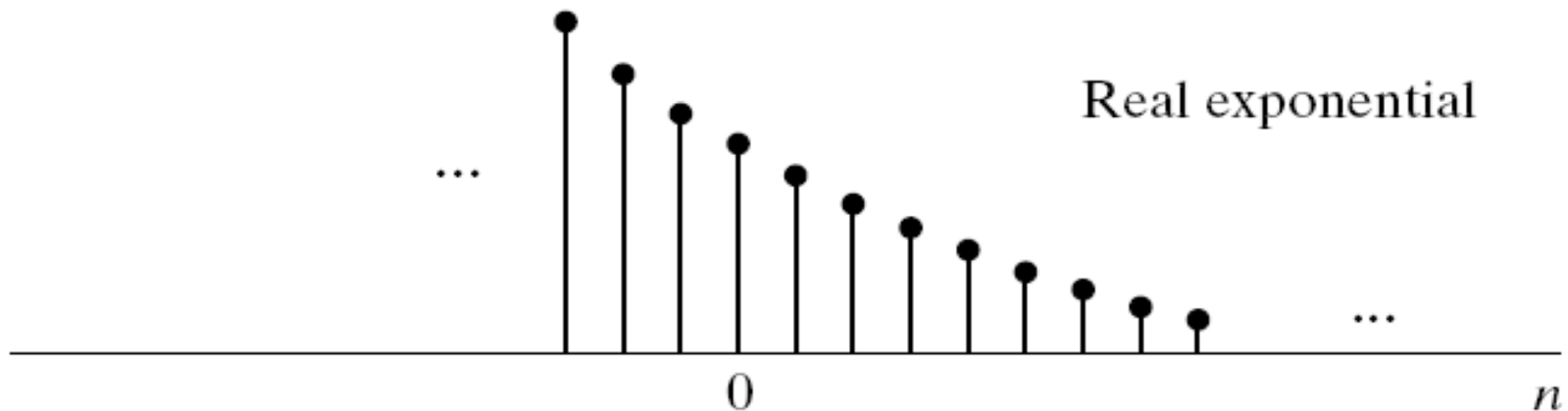
$$u[n] = \sum_{k=-\infty}^n \delta[k] \quad \left(\begin{array}{l} \sum_{k=-\infty}^n \delta[k] = \begin{cases} 0, & \text{when } n < 0 \\ 1, & \text{when } n \geq 0 \end{cases} , \\ \text{since } \delta[k] = \begin{cases} 0 & k \neq 0 \\ 1 & k = 0 \end{cases} \end{array} \right)$$

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \cdots = \sum_{k=0}^{\infty} \delta[n-k]$$

$$\delta[n] = u[n] - u[n-1]$$

• Exponential sequences $x[n] = A\alpha^n$

$$|\alpha| < 1$$



EXAMPLE

If we want an exponential sequences that is zero for $n < 0$, then

$$x[n] = \begin{cases} A\alpha^n & n \geq 0 \\ 0 & n < 0 \end{cases} \quad \text{Cumbersome}$$

$$x[n] = A\alpha^n u[n] \quad \text{simpler}$$

- Exponential Sequences

$$A = |A|e^{j\phi} \quad \alpha = |\alpha|e^{jw_0}$$

$$\begin{aligned} x[n] &= A\alpha^n = |A|e^{j\phi} |\alpha|^n e^{jw_0n} = |A||\alpha|^n e^{j(w_0n+\phi)} \\ &= |A||\alpha|^n \cos(w_0n + \phi) + j|A||\alpha|^n \sin(w_0n + \phi) \end{aligned}$$

$|\alpha| > 1$ Exponentially growing envelope

$|\alpha| < 1$ Exponentially decreasing envelope

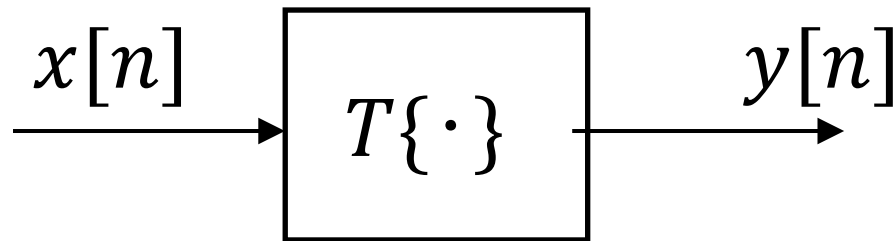
$|\alpha| = 1$ $x[n] = Ae^{jw_0n}$ is referred to
Complex Exponential Sequences

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Discrete-Time Systems is a transformation or operator that maps input sequence $x[n]$ into a unique $y[n]$

$$y[n] = T\{x[n]\},$$

$x[n], y[n]$: discrete-time signal



Discrete-Time System

- The Ideal Delay System

$$y[n] = x[n - n_d], \quad -\infty < n < \infty$$

- n_d is a positive integer $\rightarrow$ Shift the input sequence to the right by n_d samples to form the output .
- n_d is a negative integer $\rightarrow$ the system will shift the input to the left by $|n_d|$ samples.

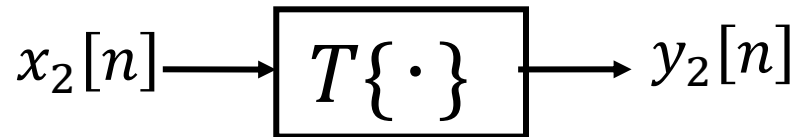
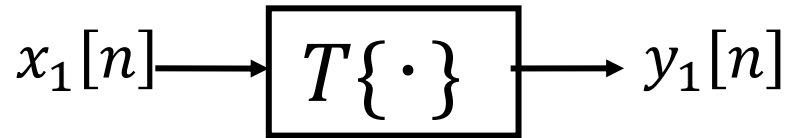
- Properties of Discrete-time systems
 - Memoryless systems:

The output $y[n]$ at every value of n depends only on the input $x[n]$ at the same value of n

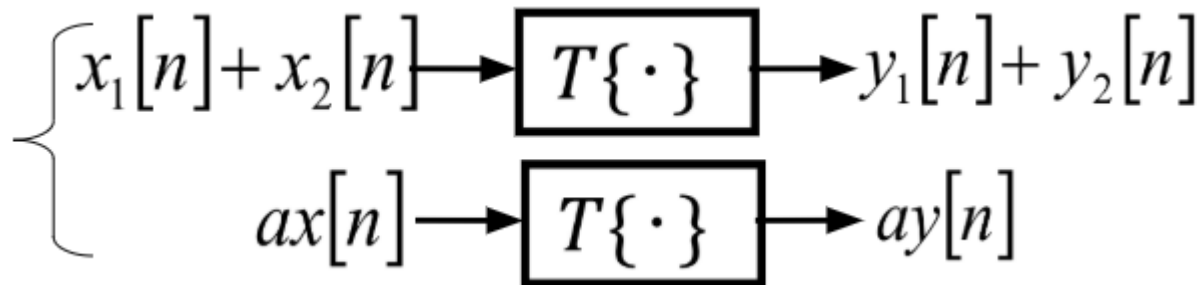
$$y[n] = (x[n])^2$$

• Linear Systems

IF



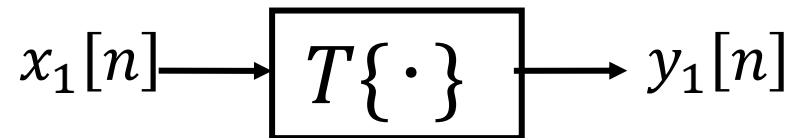
and only If:



additivity property

scaling property

- Time-Invariant Systems
 - Shift-Invariant Systems



$$x_2[n] = x_1[n - n_0] \longrightarrow T\{\cdot\} \longrightarrow y_2[n] = y_1[n - n_0]$$

•Causality

- Forward difference system is **not Causal**

$$y[n] = x[n+1] - x[n]$$

- Backward difference system is **Causal**

$$y[n] = x[n] - x[n-1]$$

• Stability

Bounded-Input Bounded-Output (**BIBO**) **Stability**: every bounded input sequence produces a bounded output sequence.

$$\text{if} \quad |x[n]| \leq B_x < \infty, \quad \text{for all } n$$

$$\text{then} \quad |y[n]| \leq B_y < \infty, \quad \text{for all } n$$

- Example for Stability or Instability

$$y[n] = (x[n])^2 \quad \text{is stable}$$

if $|x[n]| \leq B_x < \infty, \quad \text{for all } n$

then $|y[n]| \leq B_y = B_x^2 < \infty, \quad \text{for all } n$

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- Impulse response

$$\delta[n] \longrightarrow \boxed{T\{\cdot\}} \longrightarrow h[n]$$

$$\delta[n - n_0] \longrightarrow \boxed{T\{\cdot\}} \longrightarrow h[n - n_0]$$

- Representation of a general sequence as a linear combination of delayed impulses

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

- Principle of superposition

$$y[n] = T \left\{ \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \right\} = \sum_{k=-\infty}^{\infty} x[k] T \{ \delta[n-k] \}$$

$$= \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

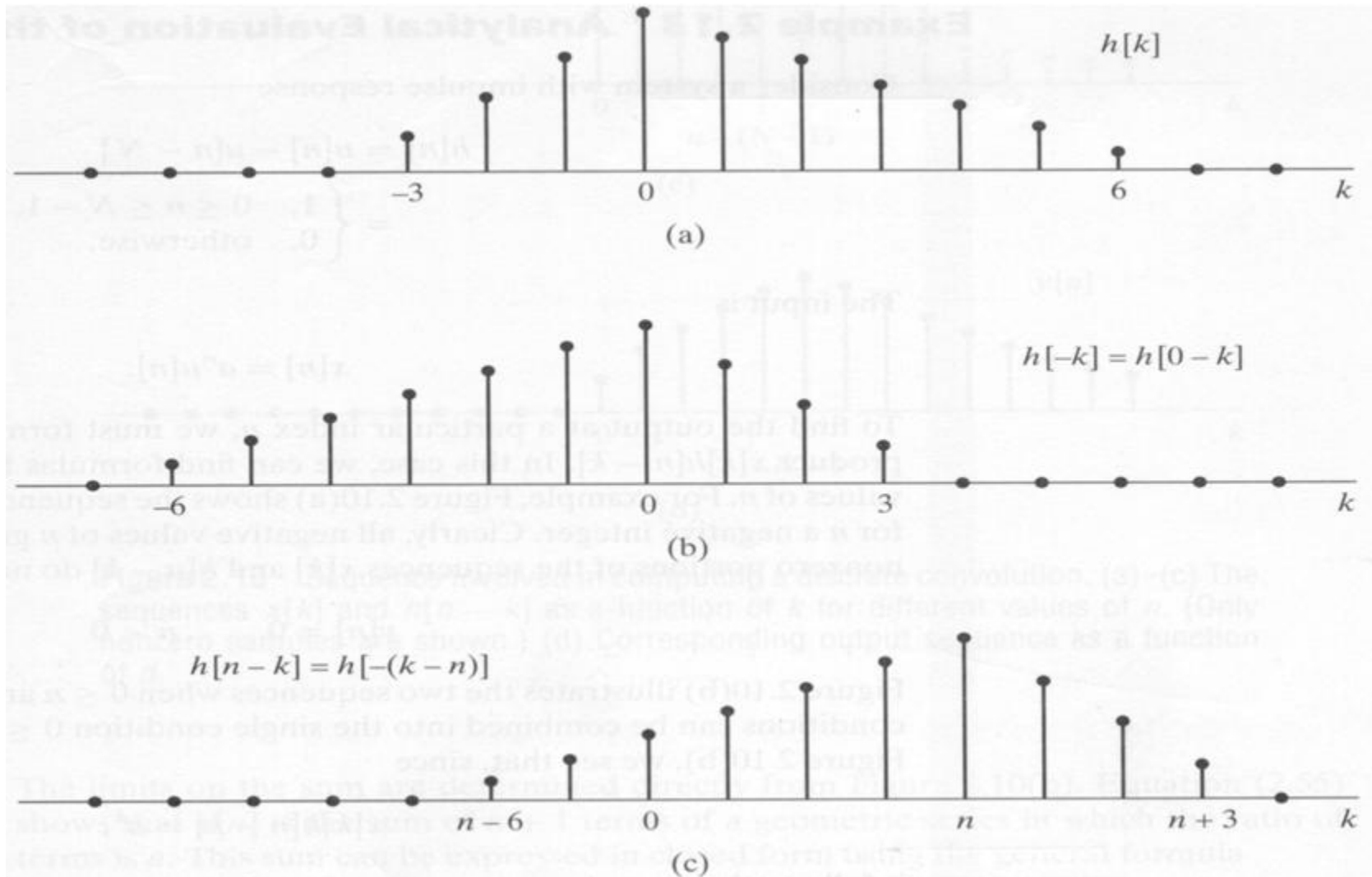
- Computation of the Convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$h[n-k] = h[-(k-n)]$$

$$h[k] \xrightarrow{\text{blue arrow}} h[-k] \xrightarrow{\text{blue arrow}}$$

- Reflecting $h[k]$ about the origin to obtain $h[-k]$
- Shifting the origin of the reflected sequence to $k=n$



- Properties of LTI Systems
 - Convolution is **commutative**

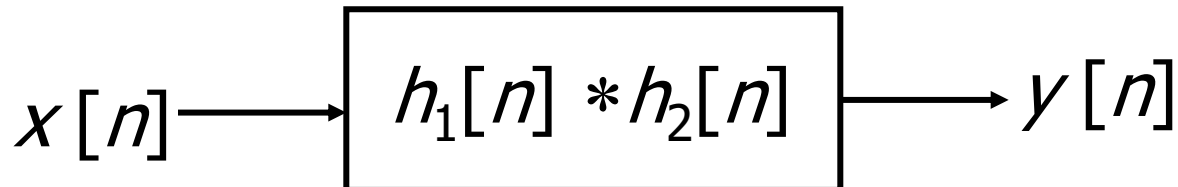
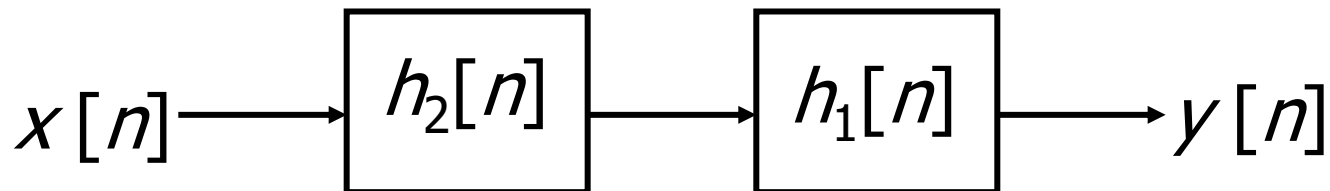
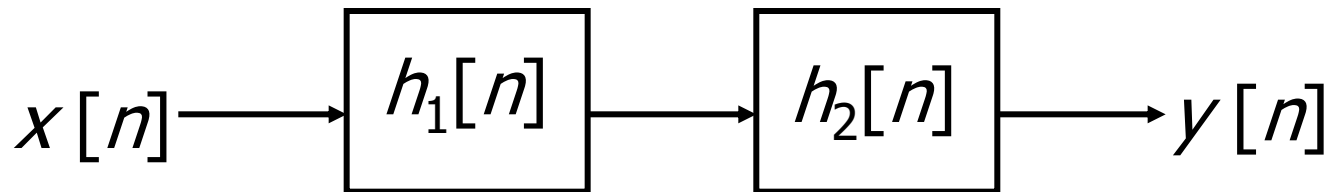
$$x[n] * h[n] = h[n] * x[n]$$

- Convolution is **distributed** over addition

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

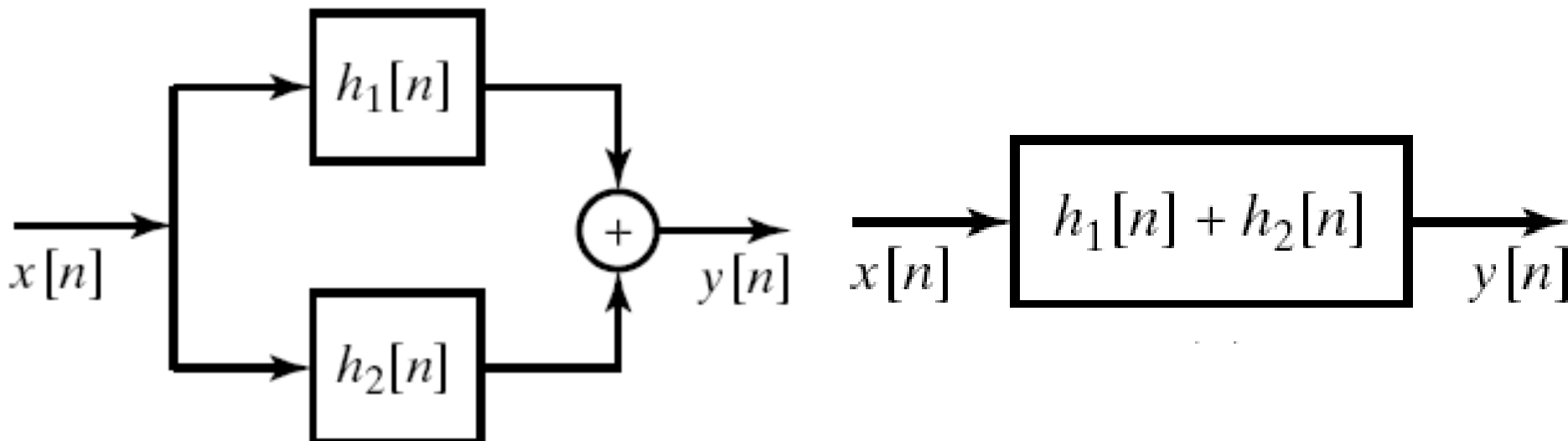
- Cascade connection of systems

$$h[n] = h_1[n] * h_2[n]$$



- Parallel connection of systems

$$h[n] = h_1[n] + h_2[n]$$



• Stability of LTI Systems

LTI system is stable if the impulse response is absolutely summable:

$$S = \sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]|$$

$$|x[n]| \leq B_x \longrightarrow |y[n]| \leq B_x \sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

Causality of LTI systems $h[n] = 0, \quad n < 0$

- Impulse response of LTI systems
 - Impulse response of Ideal Delay systems

$$h[n] = \delta[n - n_d], \quad n_d \text{ a positive fixed integer}$$

- Impulse response of Accumulator

$$h[n] = \sum_{k=-\infty}^n \delta[k] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} = u[n]$$

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- Complex exponentials are the eigenfunctions for discrete-time systems. For **LTI systems**:

$$x[n] = e^{j\omega n}, -\infty < n < \infty$$

$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} = \left(\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n}$$

$$= H(e^{j\omega}) e^{j\omega n}$$

eigenvalue



frequency response

eigenfunction



- Frequency response

$$y[n] = \left(\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n} = H(e^{j\omega}) e^{j\omega n}$$

$H(e^{j\omega})$ is called as **frequency response** of the system.

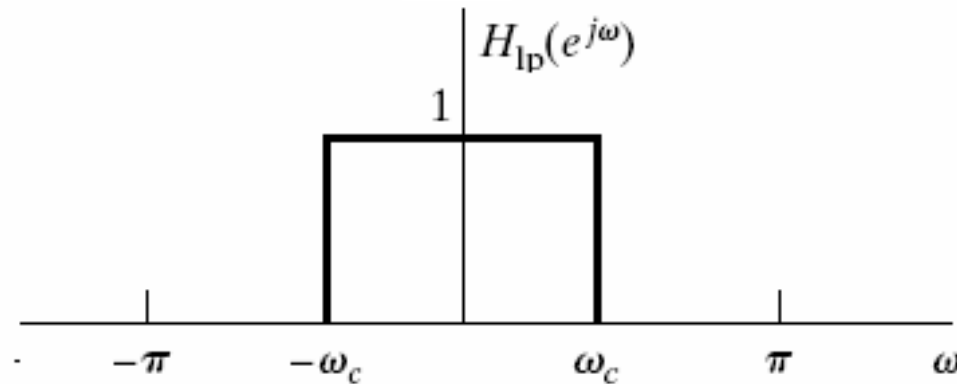
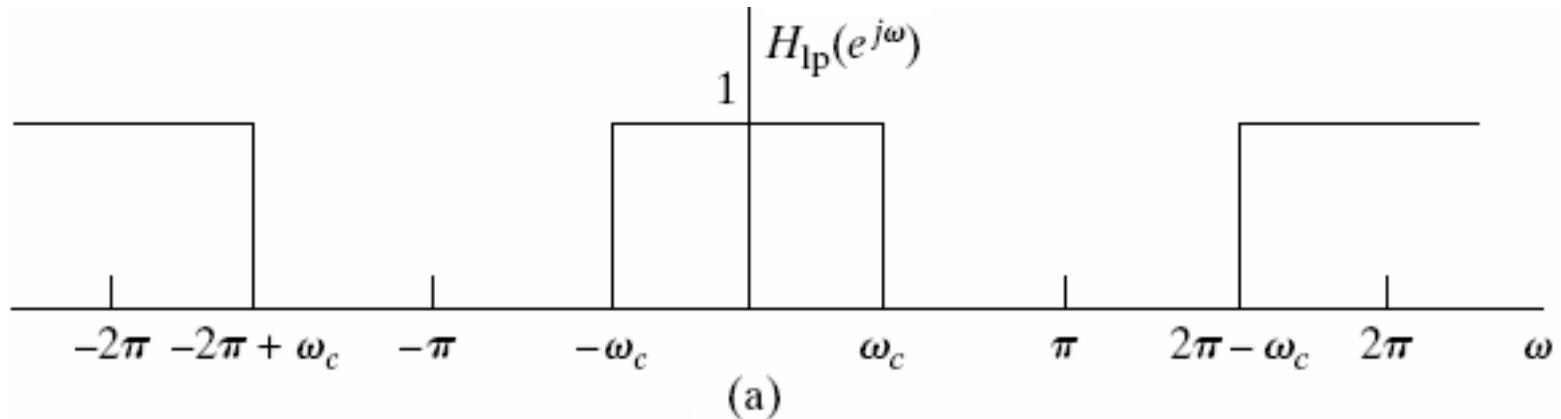
- ◆ Real part, imagine part

$$H(e^{j\omega}) = H_R(e^{j\omega}) + jH_I(e^{j\omega})$$

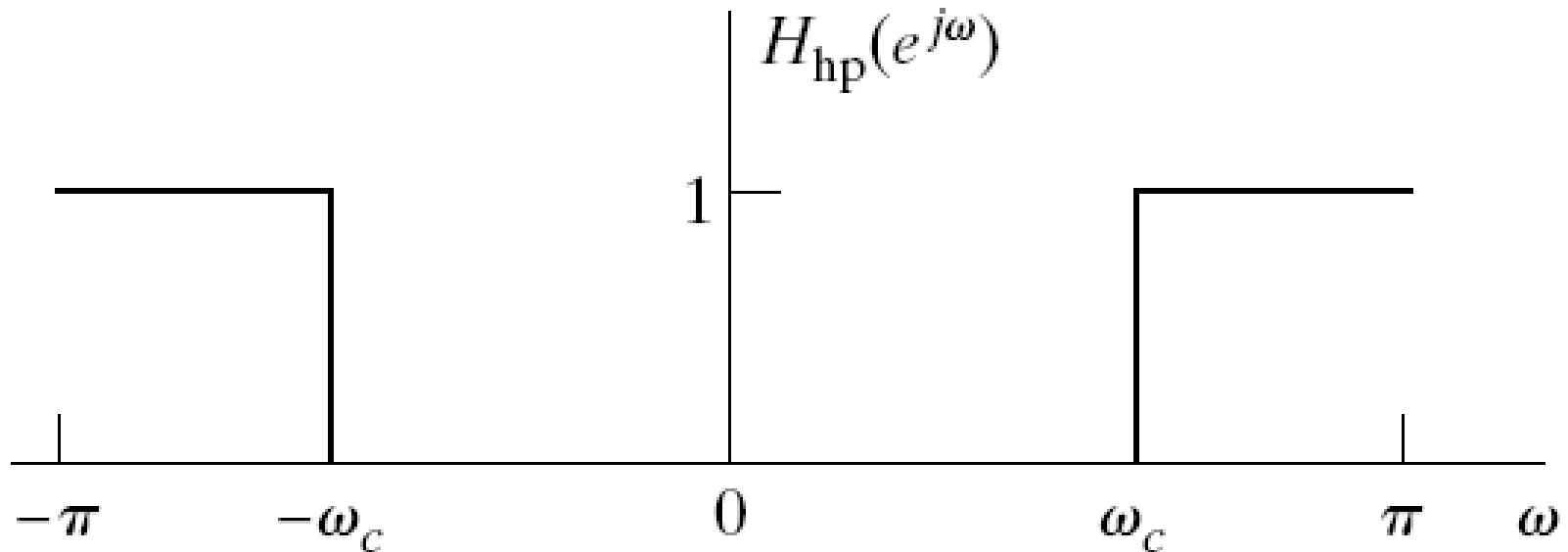
- ◆ Magnitude, phase

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}$$

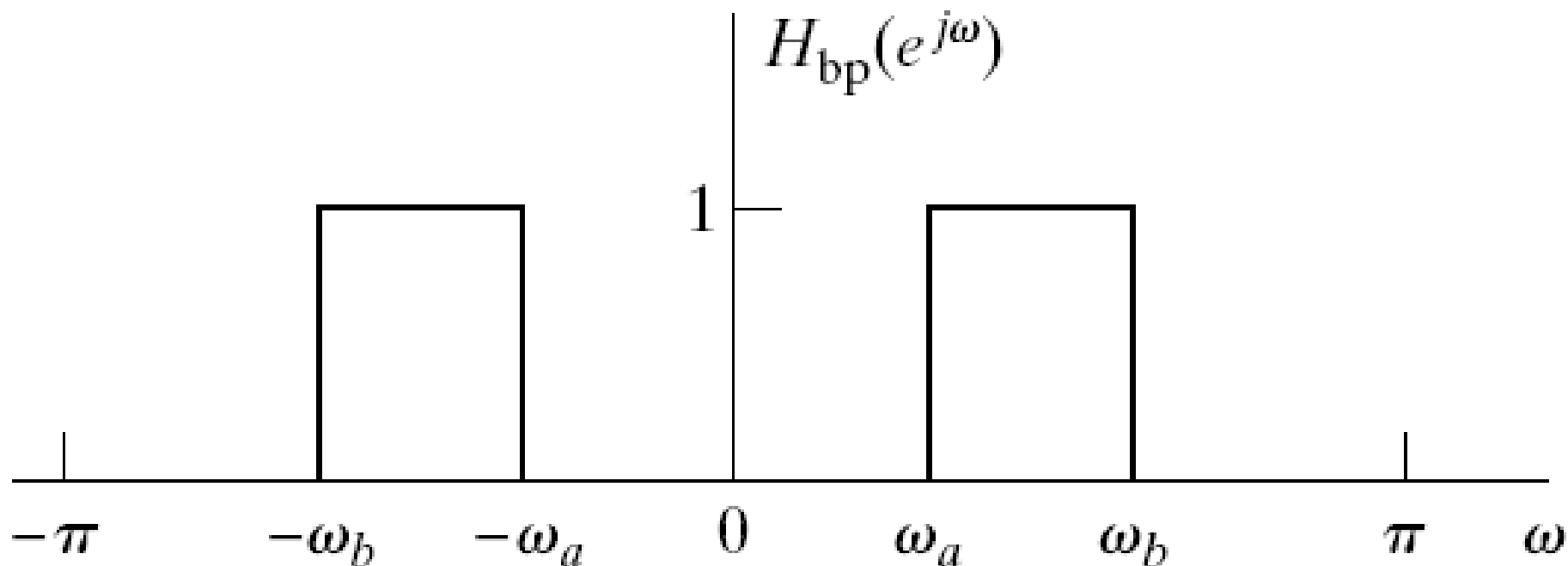
• Frequency Response of Ideal Low-pass Filter



- Frequency Response of **Ideal High-pass Filter**



- Frequency Response of Ideal Band-pass Filter



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- Discrete-Time Fourier Transform, DTFT, analyzing

$$X\left(e^{j\omega}\right)=\sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- If $x[n]$ is absolutely summable, i.e. $\sum_{n=-\infty}^{\infty} |x[n]| < \infty$ then $X\left(e^{j\omega}\right)$ exists. (Stability)

- Inverse Fourier Transform, synthesis

$$x[n]=\frac{1}{2\pi}\int_{-\pi}^{\pi} X\left(e^{j\omega}\right)e^{j\omega n}d\omega$$

- Fourier Transform

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega}) = |X(e^{j\omega})|e^{j\angle X(e^{j\omega})}$$

rectangular form polar form

$X(e^{j\omega})$: *Fourier transform,*

Fourier spectrum, spectrum

$|X(e^{j\omega})|$: *magnitude, magnitude spectrum,*

amplitude spectrum

$\angle X(e^{j\omega})$: *phase, phase spectrum*

REPRESENT THE SEQUENCES BY FOURIER TRANSFORM

- Impulse response and Frequency response
 - The frequency response of a LTI system is the Fourier transform of the impulse response.

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}$$

$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega$$

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- Conjugate-symmetric sequence

$$x_e[n] = x_e^*[-n]$$

- Conjugate-antisymmetric sequence

$$x_o[n] = -x_o^*[-n]$$

$$x[n] = x_e[n] + x_o[n]$$

$$x_e[n] = \frac{1}{2} (x[n] + x^*[-n]) = x_e^*[-n]$$

$$x_o[n] = \frac{1}{2} (x[n] - x^*[-n]) = -x_o^*[-n]$$

- Symmetry Properties of **real sequence**

- **even sequence:** a real sequence that is Conjugate-symmetric

$$x_e[n] = x_e[-n]$$

- **odd sequence:** real, Conjugate-antisymmetric

$$x_o[n] = -x_o[-n]$$

$$x[n] = x_e[n] + x_o[n]$$

real sequence:

$$x_e[n] = \frac{1}{2} (x[n] + x[-n]) = x_e[-n]$$

$$x_o[n] = \frac{1}{2} (x[n] - x[-n]) = -x_o[-n]$$

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- Linearity

$$x_1[n] \xleftrightarrow{F} X_1(e^{j\omega}) \quad x_2[n] \xleftrightarrow{F} X_2(e^{j\omega})$$

$$ax_1[n] + bx_2[n] \xleftrightarrow{F} aX_1(e^{j\omega}) + bX_2(e^{j\omega})$$

- Time shifting and frequency shifting

$$x[n] \leftrightarrow X(e^{j\omega})$$

$$x[n - n_d] \leftrightarrow e^{-j\omega n_d} X(e^{j\omega})$$

$$e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$$

• Time reversal

$$x[n] \leftrightarrow X(e^{j\omega})$$

$$x[-n] \leftrightarrow X(e^{-j\omega})$$

◆ If $x[n]$ is real, $x[-n] \leftrightarrow X^*(e^{j\omega})$

$$x[-n] = x^*[-n] \leftrightarrow X^*(e^{j\omega}) = X(e^{-j\omega})$$

$$x[n] = x^*[n] \leftrightarrow X(e^{j\omega}) = X^*(e^{-j\omega})$$

- Differentiation in Frequency

$$nx[n] \leftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

$$x[n] \leftrightarrow X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$j \frac{dX(e^{j\omega})}{d\omega} = j \sum_{n=-\infty}^{\infty} x[n] \frac{d e^{-j\omega n}}{d\omega} = \sum_{n=-\infty}^{\infty} nx[n] e^{-j\omega n}$$

• Convolution Theorem

$$x[n] \leftrightarrow X(e^{j\omega}) \quad h[n] \leftrightarrow H(e^{j\omega})$$

if
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n]$$

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

- Example: Determine the Fourier Transform of sequence

$$x[n] = a^n u[n - 5]$$

$$x_1[n] = a^n u[n] \leftrightarrow X_1(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

$$x_2[n] = x_1[n - 5] = a^{(n-5)} u[n - 5]$$

$$\leftrightarrow X_2(e^{j\omega}) = e^{-j5\omega} X_1(e^{j\omega}) = \frac{e^{-j5\omega}}{1 - ae^{-j\omega}}$$

$$x[n] = a^5 x_2[n] \leftrightarrow X(e^{j\omega}) = a^5 X_2(e^{j\omega}) = \frac{a^5 e^{-j5\omega}}{1 - ae^{-j\omega}}$$

- Example: Determine the impulse response for a difference equation:

$$y[n] - \frac{1}{2} y[n-1] = x[n] - \frac{1}{4} x[n-1]$$

- Impulse response $x[n] = \delta[n]$

$$h[n] - \frac{1}{2} h[n-1] = \delta[n] - \frac{1}{4} \delta[n-1]$$

$$H(e^{j\omega}) - \frac{1}{2} e^{-j\omega} H(e^{j\omega}) = 1 - \frac{1}{4} e^{-j\omega}$$

$$H(e^{j\omega}) = \frac{1 - \frac{1}{4} e^{-j\omega}}{1 - \frac{1}{2} e^{-j\omega}} = \frac{1}{1 - \frac{1}{2} e^{-j\omega}} - \frac{\frac{1}{4} e^{-j\omega}}{1 - \frac{1}{2} e^{-j\omega}}$$

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DISCRETE-TIME RANDOM SIGNALS

- **Deterministic**: each value of a sequence is uniquely determined by a mathematical expression, a table of data, or a rule of some type.
- **Stochastic** signal: a member of an ensemble of discrete-time signals that is characterized by a set of **probability density function**.

SUMMARY

- Define a set of basic sequence.
- Define and represent the **LTI systems** in terms of the **convolution**, **stability** and **causality**.
- Introduce the linear constant-coefficient difference equation with **initial rest conditions** for **LTI**, **causal** system.
- Recursive solution of linear constant-coefficient difference equations.

SUMMARY

- Define **frequency response** of the LTI system.
- Define **Fourier transform**.
- Introduce the properties and theorems of Fourier transform. (**Symmetry**)
- Introduce the discrete-time random signals.



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THANK YOU